How Do World Action Models Use Their Futures? Evidence from Mechanistic Interventions

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Abstract

Do world action models actually use the futures they generate? Jointly producing future observations and robot actions does not answer this question: the action may bypass the predicted future. We test future use with reachable-donor transplantation. Two native policy runs begin from the same restored simulator state and produce different futures, actions, and endpoints. We call the run being recomputed the recipient and the run supplying the alternative future the donor. Holding the recipient's observation, instruction, sampling noise, and denoising schedule fixed, we ask not merely whether its action changes, but whether it moves specifically toward the donor action. In Cosmos 3, a frozen study across 22 states and six RoboLab tasks finds predicted-future action steering of 0.992 and executed-future endpoint steering of 1.003 on a scale where 0 is the recipient and 1 is the donor. Replacing donor-run future-token keys and values with self-future counterparts removes 0.877 of the predicted-action shift, with positive action and endpoint reductions in all 22 states. Isolated robot- and object-pixel interventions, however, each have effects below 0.011. Cosmos Policy shows a related early-state effect across all ten LIBERO-Long tasks, carried mainly by visual content. These results show that both direct policies use coherent futures to condition action, but do not establish candidate search or planning over predicted outcomes.

1 Introduction

Many world action models (WAMs) generate future observations alongside robot actions [8, 14]. Joint generation, however, does not imply that the action computation uses the generated future. The policy could map the current observation and instruction directly to an action while producing a plausible future as an auxiliary output.

We distinguish four mechanistically different—and potentially coexisting—roles for future representations. First, they may be *epiphenomenal*: changing them has no causal effect on action. Second, they may provide a *nonspecific computational workspace*: activity at future-token positions is necessary, but the represented future does not determine the direction of the action change. Third, their semantic content may *parameterize action*, for example through an inverse-dynamics-like mapping from a prospective robot pose to the action that would realize it [1, 7]. Fourth, they may encode consequences that are evaluated to select among candidate actions [2, 3]. Only the fourth role entails outcome-based planning; the third establishes content-specific future use but does not, by itself, show that the model compares predicted outcomes. We study Cosmos Policy's direct-policy mode and the diffusion-generator component of Cosmos3-Nano-Policy-DROID. Neither evaluated inference procedure generates and scores multiple candidate actions before acting; Cosmos Policy's separate best-of-N planning mode is outside our study.

RIFT provides the closest causal evidence that future representations matter [22]. Masking the future read or editing values stored at future positions changes closed-loop trajectories and task success across several WAMs. Such destructive tests establish that the future channel is necessary or behav-

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

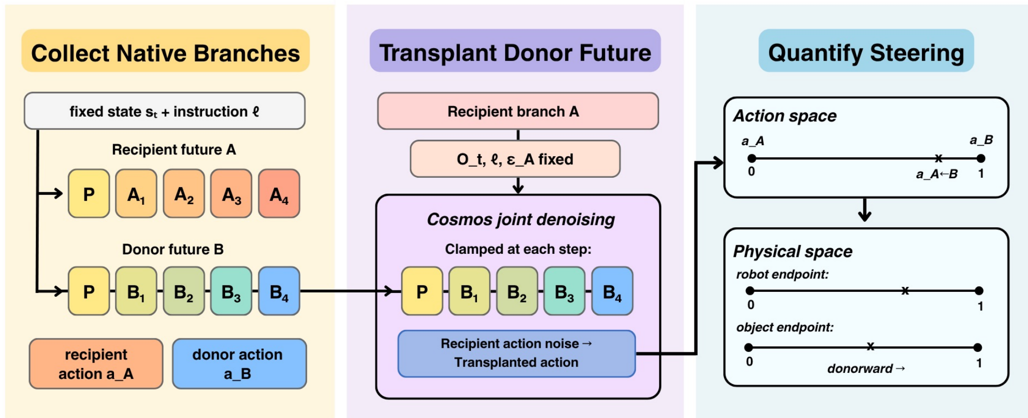

Figure 1: Reachable-donor future transplantation. (a) Two native continuations start from the same restored state. (b) The recipient is recomputed with only the donor future substituted. (c) The donor action is withheld and used afterward to score signed movement from recipient (0) toward donor (1). (d) Replacing donor-future K/V with self-future K/V tests whether that interface carries the directional shift.

iorally sensitive. They do not establish that its semantic content controls a particular alternative: a damaged representation can move an action in any direction.

Our test is directional. From the same restored state, two native policy continuations define a recipient and a donor. Both are coherent alternatives generated by the unmodified policy, and their one-chunk endpoints are reachable from that state. We insert the donor future into the recipient computation while retaining the recipient's current inputs and random draws. The model never receives the donor action; we use it only afterward to define the direction in which we score the recomputed action and its executed simulator endpoint (Figure 1). A generic perturbation need not move either quantity toward that particular donor. Directional movement therefore tests content-specific control, rather than dependence on the future channel alone.

We ask three questions: Do coherent donor futures steer behavior? Which parts of the future recreate that effect? Which future-to-action interfaces carry it? Cosmos 3 provides the strongest within-model chain: steering generalizes across 22 sampled states, isolated robot or object pixels do not reproduce it, and future-token K/V replacement suppresses action and endpoint steering in every state. Cosmos Policy supplies an earlier-model comparison whose effects are concentrated at early states and are most consistent with camera-visible robot motion. The evidence supports future-dependent action generation, but not candidate search, outcome comparison, or task-level planning.

2 Related Work

Prediction and control. Some visual robot policies generate future frames and infer actions from them; others condition action inference on predicted video representations [1, 7]. Joint video–action models instead learn future observations and controls in one model [6, 8, 10, 14, 20]. Their training objective does not determine whether a particular generated future is used at inference. UVA, for example, can omit video generation at test time [10]. Explicit visual planners answer a different question by sampling alternatives, predicting their outcomes, and comparing them with a goal-dependent score [2, 3].

Recent WAM studies separate predictive training from the use of predictions during action generation. Fast-WAM removes explicit test-time rollout while retaining competitive performance in its evaluations [21]; Faster-WAM retains a lower-cost future-conditioning route and reports improved robustness [24]. AGRA finds that visually plausible foresight need not support reliable action inference in its studied architecture [16]. RIFT directly establishes dependence on future-position values and their organization [22]. Our contribution is complementary but distinct: RIFT asks whether behavior depends on the future interface, whereas reachable-donor transplantation asks whether a coherent alternative sent through that interface predicts the direction of behavior.

Mechanistic interventions. Activation replacement tests whether an internal representation affects an output rather than merely being decodable from it [12, 18]. Path patching extends this logic to specified computational routes [5]. Conclusions can depend on the counterfactual pair and metric [23], while multi-component effects need not add when patched and unpatched routes interact [17]. We therefore treat K/V replacement as evidence for an active pathway, not a complete circuit decomposition.

3 Methods

3.1 Future transplantation and directional projection

Let saved simulator state S admit two unmodified branches A and B, with future representations F_A, F_B, action chunks $\mathbf{a}_A, \mathbf{a}_B$, and endpoints $\mathbf{e}_A, \mathbf{e}_B$ obtained by executing one action chunk. We call A the recipient—the branch whose action is recomputed—and B the donor—the branch supplying the alternative future. To recompute A, we restore the same state and observation, retain its instruction, action noise, future-noise path, solver, and denoising schedule, and replace only F_A with F_B. We repeat the intervention in the reciprocal direction, transplanting F_A into B.

For an action or endpoint $\mathbf{x} \in \{\mathbf{a}, \mathbf{e}\}$, normalized donor projection is

$$\pi_x(\hat{\mathbf{x}}; A, B) = \frac{(\hat{\mathbf{x}} - \mathbf{x}_A)^\top (\mathbf{x}_B - \mathbf{x}_A)}{\|\mathbf{x}_B - \mathbf{x}_A\|_2^2}. \tag{1}$$

The recipient is 0, the donor is 1, and 0.5 is halfway between them along the donor direction; negative values move away and values above 1 overshoot. Because the dot product retains only this signed direction, a large change perpendicular to the donor receives little projection. The primary steering contrast subtracts the corresponding self-future recomputation:

$$\Delta_x = \pi_x(\hat{\mathbf{x}}_{\text{donor}}; A, B) - \pi_x(\hat{\mathbf{x}}_{\text{self}}; A, B). \tag{2}$$

The self condition reruns the same intervention machinery with the recipient's own future, so subtraction removes any displacement due merely to recomputation. Pairs are selected using native action and endpoint separation, never intervention outcomes. Every endpoint arm is executed after restoring the exact simulator state.

3.2 Models, populations, and controls

Cosmos Policy. We evaluate the public `Cosmos-Policy-LIBERO-Predict2-2B` checkpoint in direct-policy mode on LIBERO-Long, also called LIBERO-10 [8, 11]. The frozen grid contains 30 states: ten tasks crossed with the first query, a state after three open-loop chunks, and a state after six chunks. Eight native runs determine one reciprocal donor pair per state; three future-noise draws are repeated within direction. Endpoints concatenate goal-relevant simulator coordinates with the official nine-dimensional proprioceptive endpoint.

Cosmos 3. We evaluate the 16B `Cosmos3-Nano-Policy-DROID` checkpoint on six eligible RoboLab tasks [14, 15, 19]. Its diffusion generator has 36 two-way full-attention layers, four denoising calls, 33 raw video frames mapped to nine VAE latent frames, and 32 absolute joint-position actions. The transplant targets unconditioned future-video coordinates at the final guided-sampler boundary. A frozen screen selected 24 state clusters, capped at four per task, using only native success, action and endpoint separation, and replay validity. Twenty-two completed. Two source branches ended before producing full 33-frame videos and were left missing rather than replaced. Reciprocal directions and predicted versus executed future sources are repeated measurements within state.

Controls include exact self-future recomputation, Gaussian targets matched to the donor in norm and recipient distance, a prespecified natural-future control, within-task shuffled futures, and bitwise no-op checks. Gaussian controls test whether an equally large unstructured displacement is sufficient;

the natural and shuffled controls compare the paired donor against a prespecified non-donor continuation and mismatched futures. For Cosmos 3, identity layout capture, sampler wrapping, cache record/replay, and self patches reproduce actions bitwise; sentinels confirm that the transplant does not directly write to action coordinates. Executed-future conditions encode videos produced by physically executing reachable donor actions and replay the same environment prefix before intervention. They test whether steering survives a physically realizable continuation rather than appearing only for model-predicted imagery.

3.3 Robot and object interventions

We isolate content in complementary ways. In Cosmos Policy, first-query interventions replace the wrist-camera future, primary-camera future, or future proprioception separately. A second study crosses binary robot and object states in re-rendered videos at 20 held-out states. A decoder checks whether the intended cell remains visible. A natural-pair screen separately searches policy-generated continuations for robot-divergent/object-matched and object-divergent/robot-matched futures; only four states meet its criteria, below the planned minimum of ten.

For Cosmos 3, a separately frozen factor study selected 12 population states round-robin by native object-position separation. Ten completed across all six tasks. Starting from the recipient video, an object-priority disjoint mask copies donor object pixels, donor robot pixels outside the object mask, both, or neither. This 2×2 design tests isolated pixel factors; it does not guarantee an in-distribution video or isolate a latent semantic variable.

3.4 K/V interventions

In attention, keys help determine what an action query reads, while values carry the information returned by that read. In Cosmos Policy, action queries can attend to future keys. We continuously gate future keys at prespecified block 27, with exact all-key recomputation, equal-count random-key and current-key controls, and a block-0 control. This removal changes attention normalization, so it tests whether a late future-to-action edge is active, not the information carried by a token-count-preserving route.

Cosmos 3 permits the stronger replacement test. We record future-video keys and values from a self-future run, then substitute them for donor-run future-token K/V at the frozen direct interface. Token count, token order, current-video K/V, text K/V, action K/V, observation, instruction, and noise remain fixed. Layer and pathway choices were frozen on excluded calibration tasks before the 22-state population intervention. We report *K/V replacement loss*: the reduction in donor projection after substituting self-future K/V. This is an operational intervention contrast, not a population natural indirect effect.

3.5 Statistical analysis

The independent unit is a restored simulator state, not an individual noise draw or transplant direction. We first average reciprocal directions, donors, noise draws, and process repeats within state, then form 95% percentile intervals from 10,000 bootstrap resamples of states [4]. For the common-scale condition estimates in Table 1, we reconstruct each control's projection within unit from the registered pairwise contrasts before bootstrapping; these are not differences of aggregate means. Task-balanced sensitivity analyses average states within task before resampling tasks. We prespecified 0.10 projection units—10% of the native recipient-to-donor displacement—as the smallest effect of interest for the Cosmos Policy confirmatory family and the Cosmos 3 population and factor families [9]. An interval contained within $[-0.10, 0.10]$ supports practical equivalence to zero at that scale. Prespecified primary sign-test p-values use Holm correction.

4 Results

4.1 Do coherent futures steer behavior?

The method succeeds if three things happen: inserting a donor future moves behavior toward that donor, the shift survives simulator execution, and the paired donor outperforms non-donor controls.

Table 1: Method and controls on one scale. Each row reports donor-directed projection after inserting the named future: 0 is the self-future result and 1 is the paired donor behavior. Within each model–source–outcome block, bold marks the largest estimate, always the paired donor. The paired donor exceeds the strongest available control in all five comparisons. Control projections are reconstructed within independent unit before bootstrapping.

Model	Source	Outcome	Future inserted	Projection [95% CI]
Cosmos 3	Predicted	Action	Self	0.000 (reference)
Cosmos 3	Predicted	Action	Natural control	0.559 [0.465, 0.651]
Cosmos 3	Predicted	Action	**Paired donor**	**0.992 [0.972, 1.010]**
Cosmos 3	Executed	Action	Self	0.000 (reference)
Cosmos 3	Executed	Action	Matched Gaussian	0.041 [0.003, 0.076]
Cosmos 3	Executed	Action	Natural control	0.591 [0.491, 0.687]
Cosmos 3	Executed	Action	**Paired donor**	**0.983 [0.890, 1.064]**
Cosmos 3	Executed	Endpoint	Self	0.000 (reference)
Cosmos 3	Executed	Endpoint	**Paired donor**	**1.003 [0.950, 1.050]**
Cosmos Policy	Predicted	Action	Self	0.000 (reference)
Cosmos Policy	Predicted	Action	Matched Gaussian	0.001 [−0.004, 0.006]
Cosmos Policy	Predicted	Action	Natural control	0.395 [0.269, 0.553]
Cosmos Policy	Predicted	Action	Shuffled control	0.244 [0.152, 0.347]
Cosmos Policy	Predicted	Action	**Paired donor**	**0.499 [0.336, 0.679]**
Cosmos Policy	Predicted	Endpoint	Self	0.000 (reference)
Cosmos Policy	Predicted	Endpoint	Matched Gaussian	0.003 [−0.003, 0.008]
Cosmos Policy	Predicted	Endpoint	Natural control	0.442 [0.311, 0.598]
Cosmos Policy	Predicted	Endpoint	Shuffled control	0.275 [0.185, 0.375]
Cosmos Policy	Predicted	Endpoint	**Paired donor**	**0.552 [0.387, 0.728]**

Table 1 puts the method and every applicable control on one scale, organized by model, future source, and outcome. Figure 2 shows the same comparison visually. On this scale, 0 is the self-future result and 1 is the paired donor behavior.

Cosmos 3. Across 22 completed states and all six tasks, a predicted paired-donor future produces action steering of 0.992 ($[0.972, 1.010]$; 22/22 states positive), compared with 0.559 for the natural-future control (Table 1). Executed paired-donor videos produce action steering of 0.983, compared with 0.591 for the natural control and 0.041 for the geometry-matched Gaussian control. Executing the paired-donor-conditioned action yields endpoint steering of 1.003 ($[0.950, 1.050]$; 22/22). Thus, the effect is not explained by recomputation, an equally large unstructured displacement, or merely supplying a different valid continuation. Endpoint projection shows that the donor-directed component survives simulator execution; it does not imply equality of the full endpoint vectors. Task-balanced analyses preserve these conclusions.

Cosmos Policy. At the first query, the paired donor produces action steering of 0.499 and endpoint steering of 0.552, positive in all ten tasks (Table 1). The natural control reaches 0.395 and 0.442, respectively, while the shuffled control reaches 0.244 and 0.275 and the matched Gaussian remains near zero. The natural future is the hardest control: the paired donor exceeds it by 0.103 ($[0.048, 0.161]$; 8/10) for actions and 0.110 ($[0.054, 0.166]$; 8/10) for endpoints. This rules out the weak explanation that any different tokenizer-consistent future causes the same movement. Effects are much smaller at the separately sampled later states: after three chunks, action steering is 0.031 ($[0.006, 0.066]$) and endpoint steering is 0.004 ($[-0.043, 0.038]$); after six, they are 0.027 ($[-0.010, 0.092]$) and 0.025 ($[0.005, 0.044]$). Because each time point is a different physical state, this is state dependence associated with time, not a causal estimate of temporal decay.

No screened branch pair in either policy retained a stable success/failure label across continuation seeds. We therefore interpret these results as donor-directed changes in action and one-chunk endpoint, not changes in closed-loop task completion.

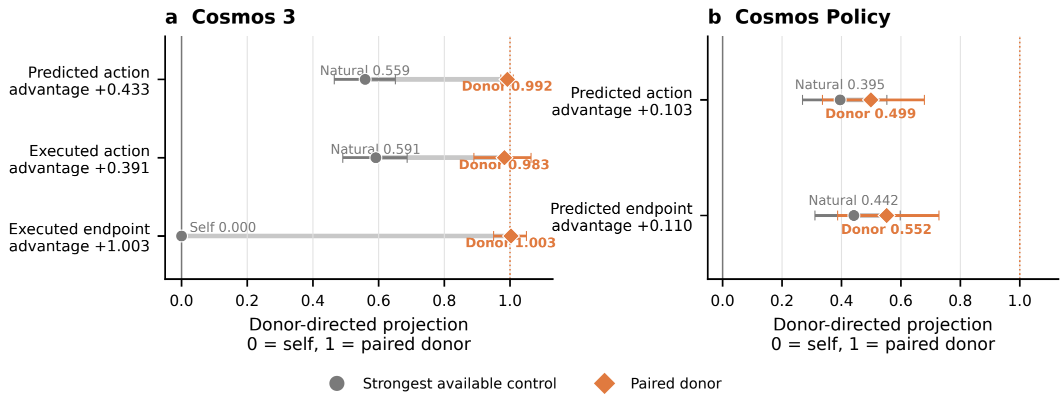

Figure 2: Paired donor versus the strongest available control, separated by model. Each row is one future-source–outcome comparison. Gray circles show the strongest control; orange diamonds show the paired donor; thin lines connect the two means. Row labels report the paired donor-minus-control advantage. Whiskers show 95% state-bootstrap intervals. All five paired-advantage intervals exclude zero; Table 1 reports the complete control set.

4.2 What future content causes steering?

The content that drives steering differs across checkpoints. Cosmos Policy responds mainly to camera-visible robot motion at early states, whereas neither isolated pixel factor recreates Cosmos 3's whole-future effect.

Cosmos Policy. Wrist-future action steering is 0.435 ($[0.284, 0.605]$), compared with 0.101 ($[0.057, 0.144]$) for the primary camera and 0.001 ($[-0.004, 0.008]$) for future proprioception (Figure 3a). In the 20-state 2×2 study, decoded futures identify the intended cell with 92.5% primary-camera and 98.75% wrist-camera accuracy. The intended object manipulation is therefore visible in the rendered futures, yet the object-state main effect is 0.00015 ($[-0.00012, 0.00049]$) for action and 0.00018 ($[-0.00040, 0.00102]$) for the executed goal-state endpoint. The available analysis does not report the robot-state main effect. Four exploratory natural pairs provide a complementary comparison: robot-motion futures yield action steering of 0.291 ($[0.060, 0.521]$) and robot-endpoint steering of 0.315 ($[0.059, 0.571]$), whereas the object-only effects are 0.021 for both action and goal endpoint, with intervals crossing zero. At these sampled early states, the pattern is most consistent with an inverse-dynamics-like role for camera-visible robot motion. It does not show that Cosmos Policy generally ignores object consequences.

Cosmos 3. Across ten factor-study states and all six tasks, every isolated-factor interval lies inside the prespecified $[-0.10, 0.10]$ equivalence region (Table 2; Figure 3b). Robot-pixel effects are 0.0022 for actions and 0.0102 for endpoints; object-pixel effects are 0.0061 and -0.0001. Neither factor alone recreates a meaningful fraction of the roughly one-unit whole-future effect. The useful signal may be distributed across the image, require robot–object coherence, depend on camera or temporal structure, or interact nonlinearly with masking and tokenization. Hybrid videos may also be off distribution. The experiment establishes insufficiency of each isolated pixel factor, not irrelevance of robot or object information and not joint necessity.

Table 2: Cosmos 3 isolated-content interventions across 10 states from six tasks. Entries are mean donor-directed projections with 95% state-bootstrap intervals; 0 denotes the recipient and 1 the paired donor. Robot and object masks are disjoint. The prespecified practical-equivalence region is $[-0.10, 0.10]$.

Donor pixels inserted	Action projection [95% CI]	Endpoint projection [95% CI]
Robot only	0.0022 [−0.0016, 0.0060]	0.0102 [−0.0067, 0.0275]
Object only	0.0061 [0.0009, 0.0119]	−0.0001 [−0.0254, 0.0239]

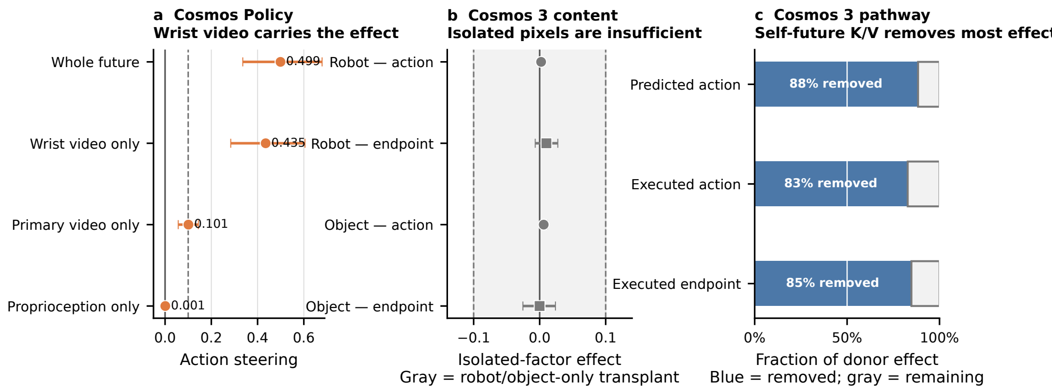

Figure 3: What carries the effect? Orange denotes donor-future steering, gray denotes isolated pixel-factor effects, and blue denotes the fraction removed by restoring self-future K/V. (a) In Cosmos Policy, the wrist-video future reproduces most of the whole-future action effect; future proprioception reproduces none. (b) In Cosmos 3, neither robot-only nor object-only pixels reproduce the whole-future effect; the shaded band is the prespecified equivalence region. (c) Replacing the donor-future K/V pathway with self-future K/V removes 83–88% of donor steering.

4.3 Which internal pathway carries the effect?

Future-token attention provides an active route from generated futures to action. Population-level K/V replacement in Cosmos 3 gives the stronger of the two pathway tests.

Cosmos Policy. Block-27 future-key removal produces a monotonic action-disruption curve: normalized L_2 change is 0.022, 0.035, 0.054, and 0.068 at gates 0.25, 0.50, 0.75, and 1.00. All-key recomputation is bitwise exact. Full removal is 0.016 more disruptive than equal-count random-key removal and 0.048 more disruptive than block-0 future-key removal; equal-count current-key removal is 0.024 larger. Executed endpoints support the same bounded conclusion, although state counts and intervals are not available for this analysis. The late future-to-action edge is active, but it is neither the only nor the largest route through the block.

Cosmos 3. The interface was selected on excluded calibration tasks and frozen before population testing. Replacing donor-run future-token K/V with self-future K/V removes 0.877 of the 0.992 predicted-action shift, 0.814 of the 0.983 executed-action shift, and 0.851 of the 1.003 endpoint shift (Table 3; Figure 3c). These losses correspond to 88%, 83%, and 85% of the respective unpatched point estimates and are positive in 22/22 states. Task-balanced point estimates are 0.877, 0.812, and 0.853; all prespecified primary sign tests remain significant after Holm correction. Because token count and attention positions remain fixed, donor-specific information travels through this future-token K/V interface. The experiment does not identify every head or feature, exclude interacting unpatched routes, or establish a complete circuit.

Table 3: Cosmos 3 future-token pathway intervention across 22 states from six tasks. Donor steering is the unpatched donor-directed projection. Replacement loss is the reduction in that projection after replacing donor-future K/V with self-future K/V; intervals are 95% state-bootstrap intervals. Relative reduction is the replacement-loss point estimate divided by unpatched donor steering.

Outcome	Donor steering	Replacement loss [95% CI]	Relative reduction
Predicted action	0.992	0.877 [0.842, 0.910]	88%
Executed action	0.983	0.814 [0.723, 0.900]	83%
Executed endpoint	1.003	0.851 [0.766, 0.922]	85%

5 Discussion and Limitations

Across both WAM generations, transplanting a coherent future predicts which alternative action the model will approach. The associated simulator endpoint moves in the same direction. RIFT established that behavior depends on an organized future interface; our result adds content-specific sufficiency by showing that a policy-native alternative predicts the signed direction of the change. The two checkpoints, however, do not appear to rely on the same content.

Cosmos Policy's early-state pattern resembles an inverse-dynamics interface conditioned on visible robot motion. That account does not transfer cleanly to Cosmos 3: strong whole-future steering coexists with negligible robot-only and object-only pixel effects. Usable content may instead depend on architecture, surrounding context, or distributed feature interactions. The pathway experiments locate where some of this influence travels. Cosmos Policy identifies an active late attention edge, while Cosmos 3 shows across 22 states that future-token K/V carries most of the donor-directed shift.

This evidence is narrower than planning. A planning claim would require evidence that the policy generates alternatives, predicts their outcomes, compares them using a goal, cost, or value, and selects accordingly. We do not test that sequence, and no screened pair retained a stable success/failure label across continuation seeds. Here, "future" and "imagined" describe generated representations, not deliberation. Our claim is that these direct policies condition action on coherent futures, not that WAMs generally plan over task consequences.

Limitations. All interventions are in simulation and cover two checkpoints, six RoboLab tasks, and LIBERO-Long. Cosmos Policy's strongest result uses one first-query state per task, while later estimates use different physical states and therefore cannot isolate elapsed time. Its natural content analysis contains only four states, and rendered hybrids may be outside the policy's native future distribution. Cosmos 3 completed 22 of 24 frozen population states and 10 of 12 factor states; the two population omissions resulted from short source videos and were not replaced. Pixel masks may miss distributed semantic features or induce nonlinear changes, so negligible isolated effects do not establish joint necessity. Key removal changes attention normalization, and K/V replacement can interact with unpatched routes. We do not demonstrate task-success mediation or recover a full circuit.

Broader impact. Mechanistic tests can help distinguish plausible-looking predictions from representations that actually control behavior. The same interfaces could also redirect robot actions or create false confidence if validated on narrow state distributions. Use beyond simulation would require broader coverage, physical safety constraints, and monitoring outside the intervention metric.

6 Conclusion

The central result is directional: we do not merely damage a model's future and observe that behavior changes; we supply a different coherent future and find that behavior follows that particular alternative. Cosmos 3 shows this across 22 sampled states, with future-token K/V carrying most of the action and endpoint shift. Cosmos Policy shows a related, more state-dependent effect dominated by visual robot-motion information, while neither isolated robot nor object pixels reproduce Cosmos 3's whole-future effect. These models use generated futures to condition action. Whether they compare alternative predicted outcomes in order to plan remains unanswered.

References

[1] Yilun Du, Sherry Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Joshua B. Tenenbaum, Dale Schuurmans, and Pieter Abbeel. Learning universal policies via text-guided video generation. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/1d5b9233ad716a43be5c0d3023cb82d0-Abstract.html.

[2] Yilun Du, Sherry Yang, Pete Florence, Fei Xia, Ayzaan Wahid, Brian Ichter, Pierre Sermanet, Tianhe Yu, Pieter Abbeel, Joshua B. Tenenbaum, Leslie Kaelbling, Andy Zeng, and Jonathan Tompson. Video language planning. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=9pKtcJcMP3.

[3] Frederik Ebert, Chelsea Finn, Sudeep Dasari, Annie Xie, Alex Lee, and Sergey Levine. Visual foresight: Model-based deep reinforcement learning for vision-based robotic control. *arXiv preprint arXiv:1812.00568*, 2018. URL https://arxiv.org/abs/1812.00568.

[4] C. A. Field and A. H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B*, 69(3):369–390, 2007. doi: 10.1111/j.1467-9868.2007.00593.x.

[5] Nicholas Goldowsky-Dill, Chris MacLeod, Lucas Sato, and Aryaman Arora. Localizing model behavior with path patching, 2023. URL https://arxiv.org/abs/2304.05969. arXiv preprint, version 2.

[6] Yanjiang Guo, Yucheng Hu, Jianke Zhang, Yen-Jen Wang, Xiaoyu Chen, Chaochao Lu, and Jianyu Chen. Prediction with action: Visual policy learning via joint denoising process. In *Advances in Neural Information Processing Systems*, volume 37, pages 112386–112410. Curran Associates, Inc., 2024. doi: 10.52202/079017-3570. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/cbe25fa0e7c7084049276888a09acc8d-Abstract-Conference.html.

[7] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 24328–24346. PMLR, 2025. URL https://proceedings.mlr.press/v267/hu25g.html.

[8] Moo Jin Kim, Yihuai Gao, Tsung-Yi Lin, Yen-Chen Lin, Yunhao Ge, Grace Lam, Percy Liang, Shuran Song, Ming-Yu Liu, Chelsea Finn, and Jinwei Gu. Cosmos Policy: Fine-tuning video models for visuomotor control and planning. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=wPEIStHxYH.

[9] Daniël Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017. doi: 10.1177/1948550617697177.

[10] Shuang Li, Yihuai Gao, Dorsa Sadigh, and Shuran Song. Unified video action model, 2025. URL https://arxiv.org/abs/2503.00200. arXiv preprint, version 3.

[11] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems*, volume 36, pages 44776–44791. Curran Associates, Inc., 2023. doi: 10.52202/075280-1939. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/8c3c666820ea055a77726d66fc7d447f-Abstract-Datasets_and_Benchmarks.html.

[12] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc., 2022. doi: 10.52202/068431-1262. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html.

[13] Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-scale simulation of household tasks for generalist robots. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024. doi: 10.15607/RSS.2024.XX.050. URL https://www.roboticsproceedings.org/rss20/p050.html.

[14] NVIDIA. Cosmos 3: Omnimodal world models for physical AI. *arXiv preprint arXiv:2606.02800*, 2026. doi: 10.48550/arXiv.2606.02800. URL https://research.nvidia.com/labs/cosmos-lab/cosmos3/technical-report.pdf.

[15] NVIDIA. Cosmos3-nano-policy-droid. Hugging Face model checkpoint and configuration, 2026. URL https://huggingface.co/nvidia/Cosmos3-Nano-Policy-DROID. Checkpoint revision 2b9f9517.

[16] Lu Qiu, Yizhuo Li, Yi Chen, Yuying Ge, Yixiao Ge, and Xihui Liu. Making foresight actionable: Repurposing representation alignment in world action models, 2026. URL https://arxiv.org/abs/2606.12217. arXiv preprint.

[17] Sankaran Vaidyanathan, David Arbour, Aaron Mueller, Scott Niekum, and David Jensen. The curse of multiple mediators: Hidden interaction effects in activation patching, 2026. URL https://arxiv.org/abs/2606.27510. arXiv preprint.

[18] Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, pages 12388–12401. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper/2020/hash/92650b2e92217715fe312e6fa7b90d82-Abstract.html.

[19] Xuning Yang, Rishit Dagli, Alex Zook, Hugo Hadfield, Ankit Goyal, Stan Birchfield, Fabio Ramos, and Jonathan Tremblay. RoboLab: A high-fidelity simulation benchmark for analysis of task generalist policies. In *Proceedings of Robotics: Science and Systems*, Sydney, Australia, July 2026. doi: 10.15607/RSS.2026.XXII.096. URL https://roboticsproceedings.org/rss22/p096.html.

[20] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, Ayaan Malik, Kyungmin Lee, William Liang, Nadun Ranawaka, Jiasheng Gu, Yinzhen Xu, Guanzhi Wang, Fengyuan Hu, Avnish Narayan, Johan Bjorck, Jing Wang, Gwanghyun Kim, Dantong Niu, Ruijie Zheng, Yuqi Xie, Jimmy Wu, Qi Wang, Ryan Julian, Danfei Xu, Yilun Du, Yevgen Chebotar, Scott Reed, Jan Kautz, Yuke Zhu, Linxi "Jim" Fan, and Joel Jang. World action models are zero-shot policies, 2026. URL https://arxiv.org/abs/2602.15922. arXiv preprint.

[21] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-WAM: Do world action models need test-time future imagination?, 2026. URL https://arxiv.org/abs/2603.16666. arXiv preprint, version 2.

[22] Chushan Zhang, Jinguang Tong, Xuesong Li, Yikai Wang, and Hongdong Li. Keep the future, drop the rollout: RIFT for world action models, 2026. URL https://arxiv.org/abs/2608.11521. arXiv preprint, version 2.

[23] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=Hf17y6u9BC.

[24] Weiheng Zhao, Haoyi Jiang, Xin Shi, Liu Liu, Fan Huang, Zhizhong Su, Wei Sui, and Xinggang Wang. Faster-WAM: Efficient inference-time future conditioning for robust world action models, 2026. URL https://arxiv.org/abs/2608.04404. arXiv preprint.

A Reproducibility Details

The Cosmos Policy experiments use public commit `18a2acca`, checkpoint revision `cb689ec0`, LIBERO, PyTorch 2.7 with CUDA 12.8, and two NVIDIA H100 80GB GPUs. The frozen design contains ten tasks, three state indices, eight native policy runs per state, one reciprocal donor pair per state, and three future-noise draws per direction. Confidence intervals use 10,000 bootstrap resamples of simulator states.

The Cosmos 3 experiments pin Cosmos commit `9aa98e5a`, Cosmos Framework commit `d4599e2e`, RoboLab commit `9db0aaf0`, and checkpoint revision `2b9f9517`. The frozen manifest has SHA-256 `46b6fe3d6abb07712d0926361587b853b2e8313a10c7faeeccb0150d030e986b`. Endpoint evaluation reconstructs the environment, restores the saved state, and replays the observation prefix. Separate controls test token layout, sampler wrapping, cache record and replay, self replacement, and the action-coordinate boundary. Complete timing and aggregate compute accounting remain to be added before submission.

B Additional Results and Boundaries

Across all 30 Cosmos Policy states, mean donor-minus-self steering is 0.186 ($[0.095, 0.290]$) for actions and 0.194 ($[0.091, 0.305]$) for endpoints. Because the three state indices may be nested within task trajectories, the first-query analysis is the cleaner confirmatory result.

The six-state RoboCasa replication is exploratory and is not pooled with LIBERO [13]. Its donor-minus-self endpoint effect is 0.0287 ($[0.0101, 0.0492]$), positive in 6/6 states; action steering is smaller and does not reliably exceed the Gaussian control. In the Cosmos 3 population, the two missing frozen states ended after 21 and 29 actions, before producing the 33-frame source video. Neither was replaced. The prespecified minimum of 20 states across five tasks and the factor-study minimum of ten states across five tasks were both met across six tasks.